

Special Relativity Notes

Four-Momentum, Doppler Effect, Lorentz Group and Spacetime Intervals

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1 Four-Momentum

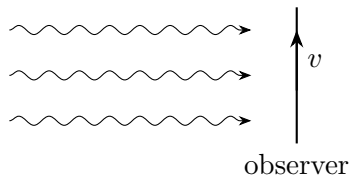
For momentum,

$$p^\mu = (p^0, \vec{p}) \quad \text{where } p^0 = \frac{E}{c}$$

$$\begin{aligned} p^\mu p_\mu &= p^0 p_0 + p^i p_i \\ &= p_0^2 - \vec{p}_i^2 \\ &= \frac{E^2}{c^2} - \vec{p}^2 \\ &= \frac{(\vec{p}^2 c^2 + m_0^2 c^4) - \vec{p}^2 c^2}{c^2} \end{aligned}$$

$$p^\mu p_\mu = m_0^2 c^2$$

2 Doppler Effect in Light



$$\nu_0 = \frac{1}{t_0}, \quad \nu = \frac{1}{t} = \frac{\sqrt{1 - v^2/c^2}}{t_0} = \nu_0 \sqrt{1 - v^2/c^2}$$

Case 1



$$T = t + \frac{vt}{c} = t \left(1 + \frac{v}{c}\right)$$

$$\nu = \frac{1}{T} = \frac{1}{t} \left(1 + \frac{v}{c}\right)^{-1} = \frac{\sqrt{1 - v^2/c^2}}{t_0 \left(1 + \frac{v}{c}\right)}$$

$$= \nu_0 \left(1 - \frac{v}{c}\right)^{1/2} \left(1 + \frac{v}{c}\right)^{1/2} \left(1 + \frac{v}{c}\right)^{-1}$$

$$\nu = \nu_0 \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}} \quad \nu < \nu_0, \quad \lambda > \lambda_0 \rightarrow \text{“Red shift”}$$

Case 2



$$T = t - \frac{vt}{c} = t \left(1 - \frac{v}{c}\right)$$

$$\nu = \frac{1}{T} = \frac{1}{t} \left(1 - \frac{v}{c}\right)^{-1} = \frac{\sqrt{1 - v^2/c^2}}{t_0 \left(1 - \frac{v}{c}\right)}$$

$$\nu = \nu_0 \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}} \quad \nu > \nu_0, \quad \lambda < \lambda_0 \rightarrow \text{“Blue shift”}$$

3 Lorentz Transformation and the Groups $SO(n)$

$$\vec{x}'_i = R_{ij} \vec{x}_j$$

In 2D, $R^T R = I$ — an **orthogonal transformation**.

- $SO(2)$: $x^2 + y^2 = r^2$ — special orthogonal group in 2 dimensions.
- $SO(3)$: $x^2 + y^2 + z^2 = r^2$ — special orthogonal group in 3 dimensions.
- $\vdots$
- $SO(n)$: $x_1^2 + x_2^2 + \cdots + x_n^2$

Lorentz transformation

$$A^\mu \rightarrow A'^\mu = \Lambda^\mu{}_\nu A^\nu \quad (\Lambda : \text{Lorentz matrix})$$

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

Lorentz scalar

$$\begin{aligned} ds^2 &= dx^\mu dx_\mu = dx'^\mu dx'_\mu = c^2 dt^2 - dx^2 - dy^2 - dz^2 & (\text{signature } +, -, -, -) \\ &= -c^2 dt^2 + dx^2 + dy^2 + dz^2 & (\text{signature } -, +, +, +) \end{aligned}$$

This preserves the quadratic form of $SO(3, 1)$.

Metric

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

(the two forms correspond to the two sign conventions).

Defining property of a Lorentz matrix

$$g_{\mu\nu} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta = g_{\alpha\beta}, \quad g^{\alpha\beta} \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta = g^{\mu\nu}$$

4 Tensors and $\det(\Lambda)$

Tensor: $A^{\mu\nu} \rightarrow$ tensor of rank 2.

$$\det(\Lambda) = \pm 1 \quad \begin{cases} +1 \Rightarrow \text{proper Lorentz transformation} \\ -1 \Rightarrow \text{improper Lorentz transformation} \end{cases}$$

Derivation of $\det(\Lambda) = \pm 1$

Write the position four-vector with $x_0 = ict$:

$$X = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad x_0 = ict$$

$$X' = \Lambda X \quad (\Lambda : \text{Lorentz transformation matrix})$$

In this form the interval is

$$ds^2 = dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2 = dX^T dX.$$

Now transform:

$$\begin{aligned} dX' &= \Lambda dX \\ ds'^2 &= dX'^T dX' \\ ds'^2 &= (\Lambda dX)^T (\Lambda dX) \\ ds'^2 &= dX^T \Lambda^T \Lambda dX \end{aligned}$$

Since the interval is invariant, $ds'^2 = ds^2 = dX^T dX$:

$$\therefore dX^T \Lambda^T \Lambda dX = dX^T dX \implies \Lambda^T \Lambda = I$$

Take the determinant:

$$\begin{aligned} |\Lambda^T \Lambda| &= |I| \\ \det(\Lambda^T) \det(\Lambda) &= 1 \\ (\det(\Lambda))^2 &= 1 \end{aligned}$$

$$\det(\Lambda) = \pm 1$$

5 Light-like, Time-like and Space-like Events

$$ds^2 = -c^2 dt^2 + d\vec{x}^2$$

$$\begin{aligned} c \, dt = dx &\implies \text{light-like event, } ds^2 = 0 \\ c \, dt > dx &\implies \text{time-like event} \\ c \, dt < dx &\implies \text{space-like event} \end{aligned}$$

Spacetime diagram

Speed of a world-line through the origin: $|v| = 0$ (the t -axis), $|v| < c$ (steeper than the light lines), $|v| = c$ (the light lines $t = \pm x/c$), and $|v| > c$ (flatter than the light lines).

